

Lecture 4: Cylindrical Shells and Method Selection

Wednesday, September 9 • Sections 2.3

• Guided Notes •

How to Use These Guided Notes: These guided notes are designed to help you organize the main ideas from each lecture but

1. we will not complete every example in class.
2. Although most definitions and theorems are included, you may need to add additional notes, examples, or explanations during class.
3. Occasionally, you will see sections labeled **Deeper Insight**. These sections provide interesting background information and help develop a deeper understanding of calculus, but they are not part of the core Calculus II content.

In today's lecture, we will continue using integration to determine the volumes of solids of revolution. The idea behind the shell method, which we will learn today, is similar to the idea behind the disk/washer method: integrating areas produces a volume. However, instead of integrating the areas of cross-sections perpendicular to the axis of rotation, we integrate the lateral surface areas of cylindrical shells that build the solid from the inside out.

Remember that

Deeper Insight

A recurring theme throughout this course will be

Integrating lengths produces areas.

A vertical or horizontal segment has a length. By accumulating infinitely many lengths, we obtain an area.

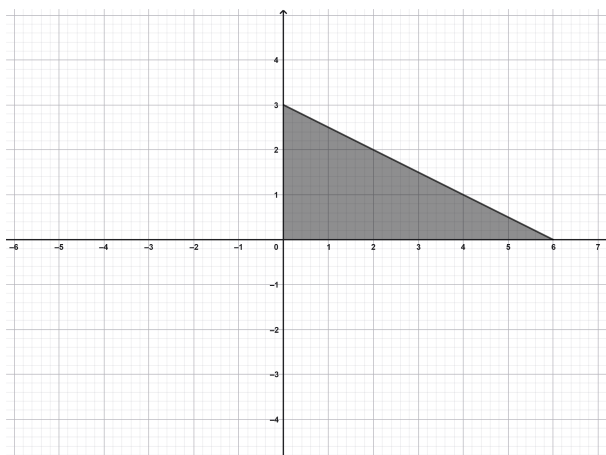
Next week, we will see that

Integrating areas produces volumes.

Understanding what quantity is being accumulated is often more important than memorizing a formula.

Example 4.1

Let us find the volume of the cone that originates when the following region is rotated about the x -axis.



1. Sketch the region. Identify the axis of rotation.
2. Draw a representative slice. Now the slice is parallel to the axis of rotation and represents a cylindrical shell, rotating the slice about the same axis of rotation, a cylindrical shell is generated.
3. Determine the variable and limits of integration: _____
4. Find radius and height of the cylindrical shells (as a function of the variable of integration):

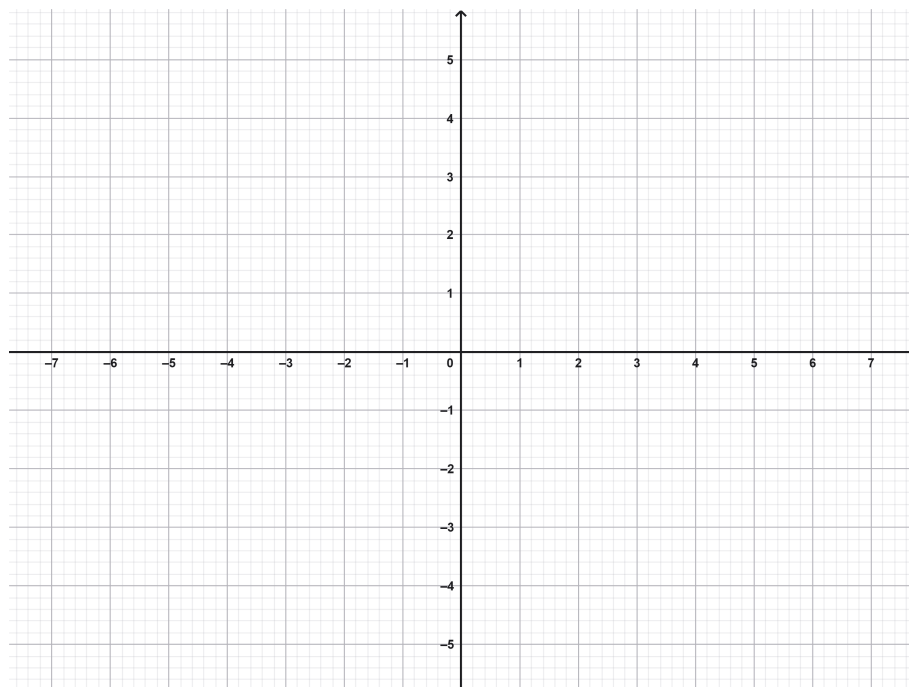
5. Write and evaluate the integral(s):

Example 4.2

Let us find the volume of the solid obtained by rotating the region bounded by the graph of $f(x) = \sqrt{x}$, the x -axis, and the lines $x = 0$ and $x = 4$ about the x -axis.

This is a desmos activity that will help you understand the solid of revolution.

1. Sketch the region. Identify the axis of rotation.



2. Draw a representative slice.
3. Determine the variable and limits of integration: _____
4. Find radius and height of the cylindrical shells (as functions of the variable of integration):

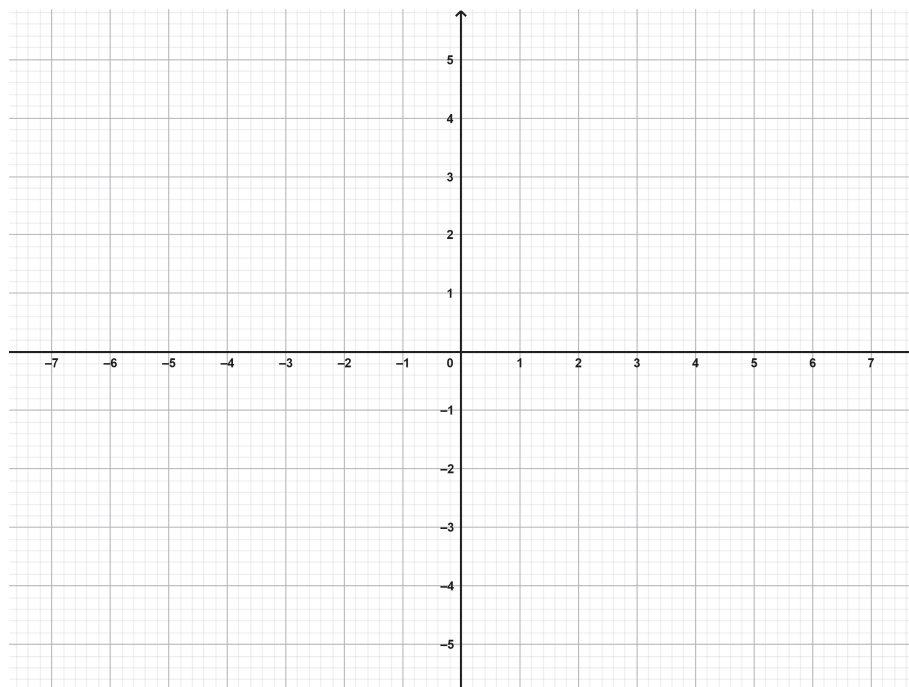
5. Write and evaluate the integral(s):

Example 4.3

Let us find the volume of the solid that originates when the same region above is rotated about the y -axis.

This is a desmos activity that will help you understand the solid of revolution.

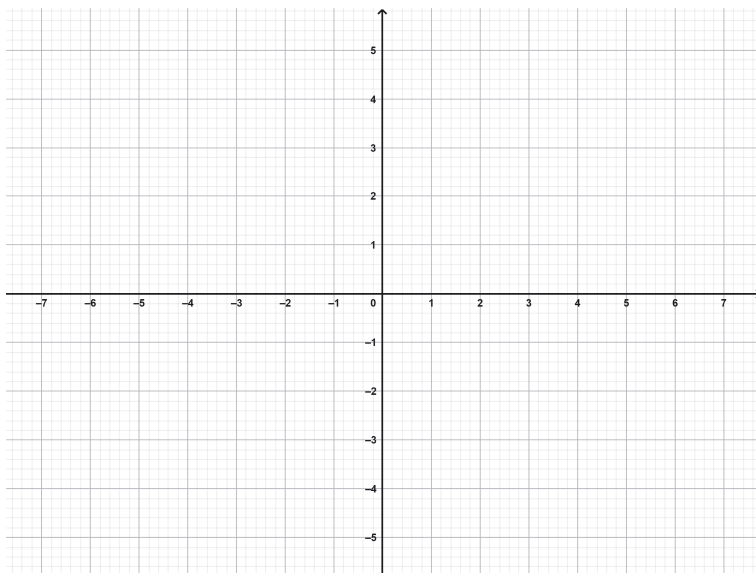
1. Sketch the region. Identify the axis of rotation.



2. Draw a representative slice.
3. Determine the variable and limits of integration: _____
4. Find radius and height (as functions of the variable of integration): _____
5. Write and evaluate the integral(s):

Example 4.4

How does the integral we previously set up change if the axis of rotation is $x = -1$?

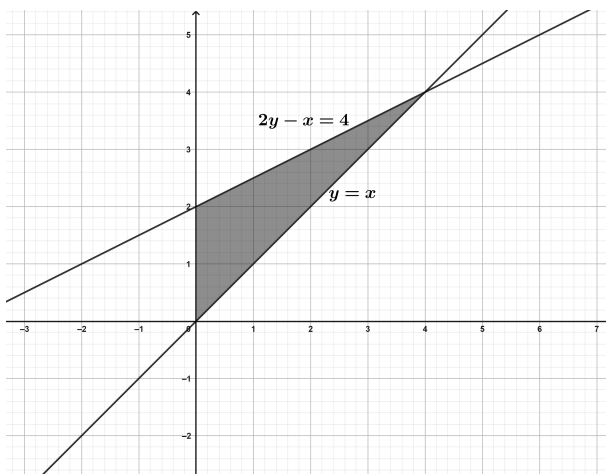


1. Sketch the region, including the axis of rotation.
2. Draw a representative slice.
3. Determine the variable and limits of integration.
4. Find the outer and inner radii.
5. Write and evaluate the integral(s).

The steps outlined in Table 1 are intended to help you set up the definite integral(s) that give the volume of a solid of revolution using the disk/washer method. These steps can be summarized as follows:

1. Sketch the region.
2. Draw a representative slice.
3. Determine the variable and limits of integration.
4. Find radius and height.
5. Write and evaluate the integral(s).

Try it yourself. Determine the volume of the solid obtained by rotating the following region about the y -axis.



1. Sketch the region. Identify the axis of rotation.
2. Draw a representative slice.
3. Determine the variable and limits of integration: _____
4. Find radius and height (as functions of the variable of integration):

5. Write and evaluate the integral(s):

Now that we know how to calculate the volume of a solid of revolution using the disks/washers and the shells methods, which method is the best one?

- Draw the region that must be rotated to generate the solid of revolution.
- Draw a slice perpendicular to the axis of rotation.
- Draw a slice parallel to the axis of rotation.

Which setup avoids solving for inverse functions? Which setup leads to fewer integrals? Which

setup gives simpler radii/heights? Choose the simpler setup.

Disk/Washer Method vs. Shell Method

Disk/Washer Method

- Representative slices are **perpendicular** to the axis of rotation.
- Integrate **cross-sectional area** A .
- Determine inner radius r and outer radius R of disks/washers.
- $A = \pi(R^2 - r^2)$.
- Volume:

$$V = \int A(x) \, dx \quad \text{or} \quad V = \int A(y) \, dy.$$

Shell Method

- Representative slices are **parallel** to the axis of rotation.
- Integrate **lateral surface area** S .
- Determine radius r and height h of cylindrical shells.
- $S = 2\pi r h$.
- Volume:

$$V = \int S(x) \, dx \quad \text{or} \quad V = \int S(y) \, dy.$$

Now, let us practice these methods. Try to avoid these common mistakes.

! Common Mistakes

- Using slices perpendicular to the axis of rotation instead of parallel to it.
- The radius is not necessarily x (or y) and the height is not necessarily $f(x) - g(x)$. The radius is the distance between the slice and the axis of rotation. The height is the distance between top and bottom boundaries (or left and right).
- Forgetting the factor 2π .
- Using the shell method but integrating with respect to the wrong variable.
- Not expressing the height in terms of the integration variable.
- Choosing shells when washers are substantially easier or viceversa.

Practice Problems

Level I problems

1. Determine the volume of the solid generated by revolving the region bounded by the graphs of $f(x) = \sqrt{x} + 1$, $g(x) = x^2 + 1$ about the x -axis.
2. Let R be the region bounded by the graph of $f(x) = e^x$ and the lines $x = 1$, $x = 2$, and $y = 0$. Choose, from the following, all true statements.
 - A. If R is rotated about the x -axis, then the disk method can be used to compute the volume of the solid without changing the independent variable.

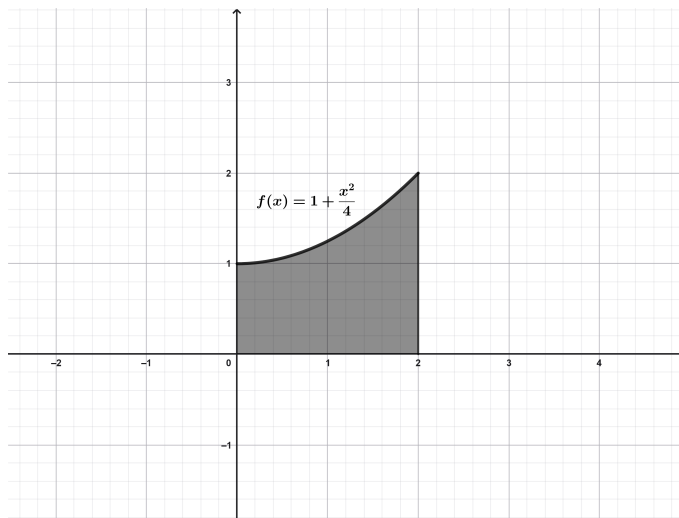
- B. If R is rotated about the y -axis, then the disk method can be used to compute the volume of the solid without changing the independent variable.
- C. If R is rotated about the x -axis, then the shell method can be used to compute the volume of the solid without changing the independent variable.
- D. If R is rotated about the y -axis, then the shell method can be used to compute the volume of the solid without changing the independent variable.
3. The region bounded by the curves $y = x^3$, $y = 8$, and $x = 0$ is revolved about the line $x = -2$ to generate a solid of revolution. Let V denote its volume. Choose **all** true statements.
- A. If the disks/washer method is used, we must integrate with respect to y to calculate V .
- B. If the shells method is used, we must integrate with respect to y to calculate V .
- C. If the disks/washer method is used, we need to evaluate a single integral to determine V .
- D. If the shells method is used, we need to evaluate a single integral to determine V .

Practice Problems

Level II problems

1. Find the volume of the following solid: the base is the region bounded from above by the graph of $f(x) = 2 - x^2$, $-1 \leq x \leq 1$ and from below by the graph of $g(x) = x^2$, $-1 \leq x \leq 1$. Parallel cross sections perpendicular to the base are circular disks.
2. Find the volume of the solid generated by rotating the region bounded by the graphs of $x = \sqrt{y}$ and $x = \frac{y}{2}$ about the line $x = 4$.
3. Calculate the volume of the solid of revolution generated by revolving the region in the figure below about the following axes of rotation.

- x -axis,
- y -axis,
- $y = 2$,
- $x = -1$,
- $x = 3$,
- $y = -1$.



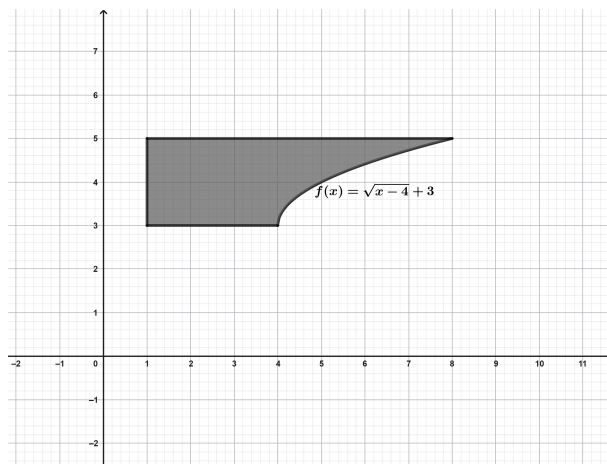
Practice Problems

Level III problems

1. Let R be the shaded region bounded by the lines

$$x = 1, y = 3, y = 5, \text{ and } f(x) = \sqrt{x - 4} + 3$$

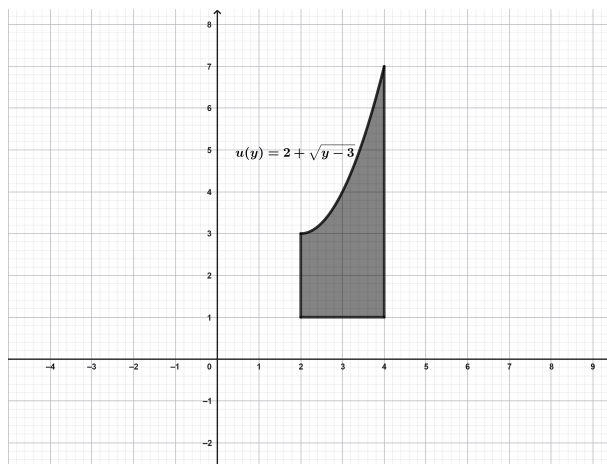
shown in the figure below. Write a SINGLE integral to calculate the volume of the solid generated when R is rotated about the x -axis.



2. Let R be the shaded region bounded by the lines

$$x = 2, x = 4, y = 1, \text{ and } u(y) = 2 + \sqrt{y - 3}$$

shown in the figure below. Write a SINGLE integral to calculate the volume of the solid generated when R is rotated about the y -axis.



After this lesson I can:

- ☐ Construct shell integrals using radius, height, and thickness.
- ☐ Compare disk/washer and shell setups for the same solid.
- ☐ Match shell orientation to the variable of integration.
- ☐ Select an efficient method and justify the choice.

Muddiest Point: What concept from today's lesson is still unclear?

Step	What to do
Sketch the region and identify the axis of rotation	1. Draw the curves that enclose the region to be rotated. 2. Shade the region to be rotated. 3. Mark the axis of rotation. 4. Identify the coordinates of the leftmost, rightmost, lowest and highest points of the region. Let a and b be the x-coordinates of the leftmost and rightmost points. Let c and d be the y-coordinates of the lowest and highest points.
Draw a representative slice	When using the shell method, the slices are parallel to the axis of rotation. Each slice intersects the boundary of the region at two points, P and Q. Because the slice is parallel to the axis of rotation, the points P and Q are at the same distance from the axis of rotation.
Determine the variable and limits of integration	1. Vertical slices: integrate with respect to x. The limits of integration are a and b (identified in step 1). 2. Horizontal slices: integrate with respect to y. The limits of integration are c and d (identified in step 1). Remember: For the shell method, the variable of integration corresponds to the axis perpendicular to the axis of rotation.
Find the function to be integrated	For each slice, identify the outer radius and height of the cylindrical shell. • The radius r of the cylindrical shell is the distance from the representative slice to the axis of rotation, it is the distance from either P or Q to the shell since, as mentioned in step 2, the distance from the axis of rotation to both P and Q is the same. • The height h of the cylindrical shell is the distance between P and Q. The area of a cylindrical shell $S = 2\pi r h$. Vertical slices: $S(x) = 2\pi r(x) h(x)$, $x \in [a, b]$. Horizontal slices: $S(y) = 2\pi r(y) h(y)$, $y \in [c, d]$. Remember: The radius of the shell is always the distance from the axis of rotation to the representative slice. The height of the shell is the distance between the points P and Q, where the slice intersects the boundary of the region.
Write and evaluate the integral	Integrate the lateral surface area of the shells over the appropriate interval. $V = \int_a^b S(x) dx \quad \text{or} \quad V = \int_c^d S(y) dy.$ Evaluate the integral to obtain the volume of the solid.

Table 1: Setting up the definite integral for the shells method